

Charles M Smith

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An undergraduate student in Mechanical Engineering committed to researching and exploring a future in robotic systems. Eager to push the boundary of research to develop solutions and change our understanding.

EDUCATION

- **Virginia Commonwealth University** Richmond, Virginia
Bachelor of Science in Mechanical Engineering *August 2023 – May. 2028*

EXPERIENCE

- **University of Virginia** Charlottesville, Virginia
Robotics Research Intern *June 2022 - August 2022*
 - **Immersive Driving Simulator:**
 - * Repurposed the cabin interior of a vehicle fixing lights, radio, and interior fans to replicate the environment of a vehicles cabin.
 - * Developed a dashboard and LCD to show relevant driving data including speed, gas, RPM, and gear along with other display lights.
 - * Interpreted telemetry data from computer simulation to replicate vehicle motion on a MOOG motion base while maintaining safety and accuracy.
 - **ROS Bots:**
 - * Worked on educational robots utilizing ROS for experimental robotics course.
 - * Analyzed each robot by testing programs and the computers, testing continuity in wiring, and measure voltage of battery cells to determine needed repairs.
 - * Diagnosed and resolved motor controller failures, improving platform reliability for instructional use
 - * Developed and deployed test programs to validate motor response and sensor feedback ensuring reliable operation for instructional and experimental use.
 - * Maintained and monitored Lipo battery health ensuring safer operating conditions.
 - **Industrial Grade Robots:**
 - * Made repairs to industrial ground robots replacing motors and rerunning cables.
 - * Replaced the axles on a 4 wheel steering robot using hardened steel to reduce bending.
 - * Assisted in integrating a Clearpath Husky mobile base with a Universal Robots manipulator, supporting experiments in mobile manipulation and coordinated motion.
- **Lowe's Home Improvement** Richmond, Virginia
Front End Fulfillment Associate *August 2024 - present*
 - **Operating Equipment:** Trained to operate counter balance forklifts, order picker forklifts, and reach trucks for loading and unloading merchandise.
 - **Sales:** Worked with customers to meet their needs with high cost purchases.

PROJECTS

- **Home Robotics (August 2022 – present):** Designed and built a custom ground robot from concept to deployment, including CAD modeling of mechanical and electronic components, fabrication of custom parts, and full system integration. Conceptualized and created a tank drive robot with a mounted Kinect camera that can be used for 3D mapping of its environment for robot world interaction.
- **Polyurethane Replication (April 2024 - present):** Optimizing a method to replicate rock textures for indoor rock climbing at a reduced cost from current methods. Reducing surface cavities by developing means to seal the surface of a mold for direct physical casting into negatives.

SKILLS

- **Languages/Libraries:** MATLAB, Java, Python (NumPy), L^AT_EX, HTML
- **Tools:** ROS, Ubuntu, Git, SOLIDWORKS, Manim
- **Hardware:** Arduino, Raspberry Pi, NVIDIA Jetson
- **Hard Skills:** Heavy equipment operation, machining, 3D printing